

Sapien-Photobomber Pipeline Report

Goal-To detect and automatically identify intended subjects in a photograph and flag photobombers for removal.

Architecture choices

1. YOLOv8m -> person class (classes[0])

Why? It handles varied scales, partial body occlusions, and varied human poses

2. Heuristic Signal Fusion -> deterministic normalised scoring model

Why? On smaller datasets, overfitting can be largely seen, using mathematical formulas for classification normalises the accuracy of the prediction.

Formulas used:

Relative Scale Score (S_{scale} — Weight: 0.45):

- Formula: $\text{Area}_i / \max(\text{Area})$
- Why? Photographic composition naturally places primary subjects closer to the lens. Background photobombers and passersby subtend smaller solid angles, producing drastically smaller 2D bounding boxes.

Center / Framing Proximity (S_{center} — Weight: 0.20):

- Formula: $1.0 - (d_{\text{center}} / d_{\text{max}})$
- Why? Photographers center or align key subjects along primary axes (Rule of Thirds). Incidental bystanders predominantly occupy the lateral periphery.

Baseline Depth Proxy (S_{base} — Weight: 0.20):

- Formula: y_2 / H_{image}
- Why? Under standard camera perspective projections, the lower boundary of an upright human bounding box serves as a monotonic proxy for ground-plane distance from the camera.

Sharpness / Optical Focus (S_{sharp} — Weight: 0.15):

- Formula: $\text{Var}(\text{Laplacian}(\text{Crop})) / \max(\text{Var})$
- Why? Portrait lenses and mobile computational portrait modes create a shallow depth-of-field (bokeh effect), leaving intentional subjects sharp while blurring background bystanders.

Results and Limitations

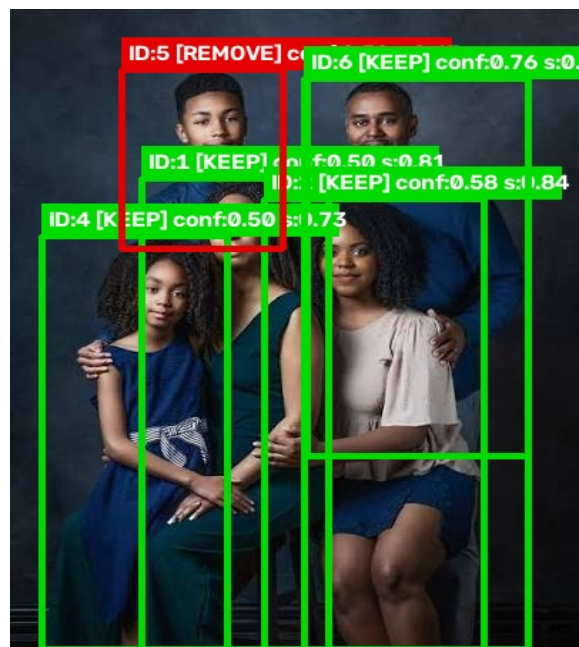
```
===== EVALUATION METRICS (Class: 'remove') =====  
True Positives (TP) : 35  
False Positives (FP): 19  
False Negatives (FN): 21  
Precision           : 0.6481  
Recall              : 0.6250  
F1-Score            : 0.6364
```

1. **False Positives**-19 cases -> Main subjects flagged as remove
 - Peripheral group members
 - Seated or receding subjects
 - High frequency background noise

Ex:



->



2. **False negatives**- 21 cases -> Photobombers flagged as keep
 - Close range photobombers
 - Centrally aligned background passerby

Ex:



Ideas for improvement

1. Detection based on facial orientation- use head-pose models like Media-pipe mesh to estimate photobombers based on altered gaze or facial position
2. Replacing heuristic formulas with metric depth estimation frameworks like Depth-Anything-v2 which completely isolates seated subjects from background noise
3. Polygonal masks- move from bounding box classification to a segmentation mask to accurately draw the main subjects
4. Graph-based -> Density-based clustering- treats the dominant group as one unit, this ensures that peripheral members do not get singled out as photobombers